AITS
ALL INDIA TEST SERIES

Dropper JEE Main

AJPM10/09 **Test- 09**

DURATION : 180 Minutes **DATE : 02/03/2025** **M.MARKS : 300**

ANSWER KEY

PHYSICS

1. (3)
2. (4)
3. (1)
4. (3)
5. (4)
6. (3)
7. (2)
8. (3)
9. (4)
10. (1)
11. (3)
12. (1)
13. (4)
14. (3)
15. (2)
16. (2)
17. (1)
18. (2)
19. (2)
20. (4)
21. (3)
22. (4)
23. (13)
24. (50)
25. (4)

CHEMISTRY

26. (2)
27. (4)
28. (2)
29. (3)
30. (2)
31. (2)
32. (4)
33. (3)
34. (2)
35. (1)
36. (1)
37. (1)
38. (3)
39. (3)
40. (3)
41. (4)
42. (2)
43. (2)
44. (1)
45. (2)
46. (3)
47. (50)
48. (92)
49. (4)
50. (3)

MATHEMATICS

51. (1)
52. (2)
53. (1)
54. (3)
55. (1)
56. (3)
57. (3)
58. (2)
59. (3)
60. (3)
61. (3)
62. (2)
63. (2)
64. (4)
65. (4)
66. (1)
67. (3)
68. (1)
69. (1)
70. (1)
71. (171)
72. (8)
73. (5)
74. (1)
75. (5)

SECTION-I (PHYSICS)

1. **(3)**

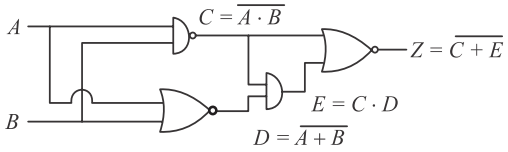
b has the dimension of volume, $[b] = [V]$

$\because \left[\frac{a}{V^2}\right] = [P]$

$\frac{a}{b} = ML^2T^{-2}$

[1 Feb, 2023 (Shift-I)]

2. **(4)**



A	B	Z
1	0	0
0	0	0
1	1	1
0	1	0

[2 Sep, 2020 (Shift-II)]

3. **(1)**

$v^2 = 2 \times 2gh$

$v = 2\sqrt{hg}$

$0 = 4hg - 2h_1 g$

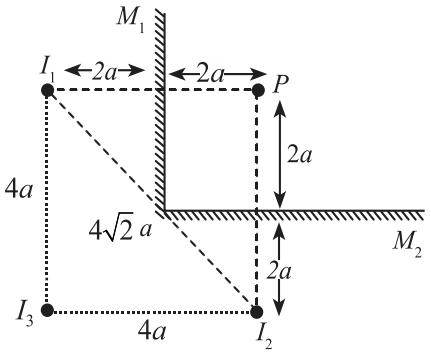
$h_1 = 2h$

$H = 2h + h = 3h$

$V_{\text{final}} = \sqrt{6hg}$

[5 Sep, 2020 (Shift-I)]

4. **(3)**

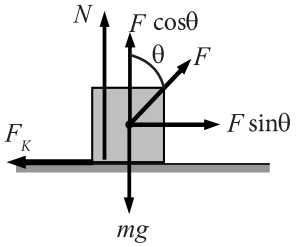


The shortest distance between the image thus formed = $4a$

[31 Aug, 2021 (Shift-I)]

5. **(4)**

From the free body diagram given below



$N = mg - F\cos\theta$ .....(i)

$ma = F\sin\theta - \mu_k N$ .....(ii)

$ma = F\sin\theta - \mu_k(mg - F\cos\theta)$

$\Rightarrow \quad a = \frac{F}{m}\sin\theta - \mu_k\left(g - \frac{F}{m}\cos\theta\right)$

[16 March, 2021 (Shift-I)]

6. **(3)**

The fringe separation $\beta = \frac{\lambda D}{d} = \frac{600 \times 10^{-9} \times 1}{1 \times 10^{-3}}$

= 0.60 mm

[26 Feb, 2021 (Shift-I)]

7. **(2)**

$\alpha = \frac{d\omega}{dt}$

$\Rightarrow \int_{10}^{\omega} d\omega = \int_0^t \alpha dt$

$\Rightarrow [\omega]_{10}^{\omega} = \int_0^t \alpha dt$

$\Rightarrow \omega - 10 = \int_0^t \left(3t^3 - t^2 + 2t\right)dt$

$\omega = \frac{3}{4}t^4 - \frac{t^3}{3} + t^2 + 10$

$\frac{d\theta}{dt} = \frac{3}{4}t^4 - \frac{t^3}{3} + t^2 + 10$

$\Rightarrow \int_4^{\theta} d\theta = \int_0^t \left(\frac{3}{4}t^4 - \frac{t^3}{3} + t^2 + 10\right)dt$

$\theta = \frac{3}{20}t^5 - \frac{t^4}{12} + \frac{t^3}{3} + 10t + 4$

[28 June, 2022 (Shift-II)]

8. **(3)**

Moment of inertia of uniform disc. $I = \frac{MR^2}{2}$

$I = \frac{\left[\left(\rho\pi R^2 t\right)\right]R^2}{2}$

[Where ρ is uniform density and *t* is thickness

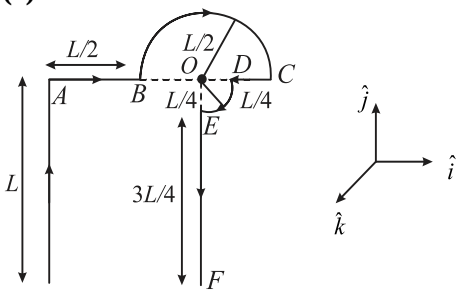
$I_1 \propto R_1^4$ & $I_2 \propto R_2^4$

$\frac{I_1}{I_2} = \left(\frac{R_1}{R_2}\right)^4 = \frac{1}{\alpha^2} = \frac{1}{16}$

$\alpha = 4$

[4 Sep, 2020 (Shift-II)]

9. (4)



$\vec{B}_{net} = \vec{B}_{AB} + \vec{B}_{BC} + \vec{B}_{CD} + \vec{B}_{DE} + \vec{B}_{EF}$

$\vec{B}_{BC} = \left(\frac{\mu_0 I \times 2}{2L} \times \frac{1}{2}\right)\left(-\hat{k}\right)$

$\vec{B}_{DE} = \left(\frac{\mu_0 I \times 4}{2L} \times \frac{1}{4}\right)\left(-\hat{k}\right)$

$\vec{B}_{net} = \frac{\mu_0 I}{L}\left(-\hat{k}\right)$

[24 June, 2022 (Shift-I)]

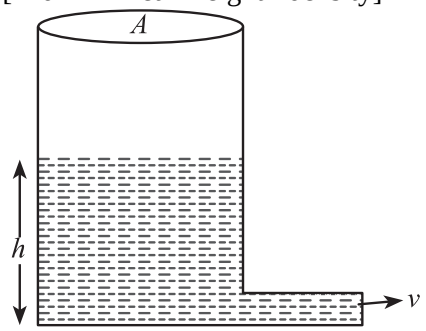
10. (1)

Velocity of efflux, $v = \sqrt{2gh}$

Thrust on the hole

$F = v\frac{dm}{dt} = v(a\rho v) = v^2 a\rho = 2gha\rho$

[$\because$ dm = Area ×height ×density]



Normal reaction on the hole, $R = mg = Ah\rho g$

Therefore, friction force, $f_r = \mu R = \mu Ah\rho g$

$F - f_r = ma_C$ (a_C = acceleration of cylinder)

$2gha\rho - \mu g\rho Ah = \rho\, Ah\, a_C$.

$\Rightarrow \quad a_C = g\left(\frac{2a}{A} - \mu\right)$

[27 July, 2021 (Shift-I)]

11. (3)

Density of molecules of gas, $n(r) = n_0 e^{-\alpha r^4}$

$\therefore$ Total number of molecule

$= \int_0^\infty n(r)dV = \int_0^\infty n_0 e^{-\alpha r^3} 4\pi r^2 dr = \frac{4\pi n_0}{3\alpha}$

[12 April, 2019 (Shift-II)]

12. (1)

A. Stopping potential depends on both frequency of light and work function.

B. Saturation current ∝ intensity of light

C. Maximum KE depends on frequency

D. Photoelectric effect is explained using particle theory

[24 Jan, 2023 (Shift-I)]

13. (4)

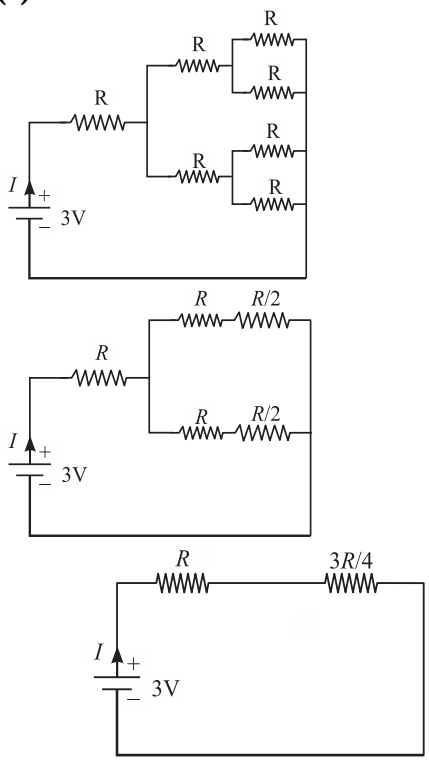
$\therefore$ Flux through curve surface of hemisphere

$\phi = \frac{q}{2\varepsilon_0}$

Note: Flux through flat surface should be zero.

[27 June, 2022 (Shift-II)]

14. (3)



Current, $I = \frac{12}{7R}$

On comparing above result with $I = \frac{12}{7}$ A.

we get, a = 12

[28 June, 2022 (Shift-II)]

15. (2)

$\because \quad \phi = BA$

$\because \quad \phi_{net} = \mu_r \mu_0 \frac{N}{L} I \times A \times N$

Thus, $\mu_r = 0.625$

[31 Jan, 2023 (Shift-I)]

16. (2)

$\frac{d\phi}{dt} > 0$ (initially)

When magnet passes through the centre region of loop, $\frac{d\phi}{dt} =$ constant

$\frac{d\phi}{dt} < 0$ (later)

[27 Aug, 2021 (Shift-I)]

17. (1)

$Q_f = CV_f$

$V = \frac{Q_f}{C} = \frac{5}{1\times10^{-6}} = 5\times10^{-6}\,\text{V}$

Since, $Q \propto V$ therefore would be a straight line passing through origin

[25 July, 2022 (Shift-I)]

18. (2)

$\hat{E}\times\hat{B}$ must be direction of propagation.

[28 June, 2022 (Shift-II)]

19. (2)

$F_1 = \frac{GMm}{(2R)^2} = \frac{GMm}{4R^2}$ …(i)

$F_2 = \frac{GMm}{(2R)^2} - \frac{G\frac{M}{8}m}{\left(\frac{3R}{2}\right)^2} = \frac{GMm}{R^2}\left(\frac{1}{4}-\frac{1}{18}\right)$

$\frac{F_1}{F_2} = \frac{9}{7}$

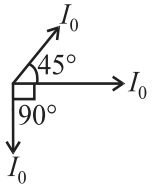
[25 Feb, 2021 (Shift-I)]

20. (4)

The second member of the Paschen series, second member of Balmer series & the series limit of Lyman series.

[24 Feb, 2021 (Shift-I)]

21. (3)



$I_{res} = 2I_0 + I_0 \Rightarrow 3I_0$

[9 Jan, 2020 (Shift-I)]

22. (4)

Slope of the curve $= \frac{\Delta L/W}{L} = \frac{\Delta L/L}{W} = \frac{1}{YA}$

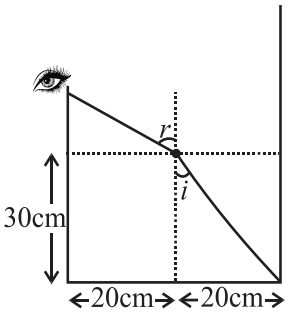
$Y = \frac{1}{(\text{Slope})\times A}$

$Y = \frac{1}{(0.25\times10^{-5})\times(1\times10^{-6})}$

$Y = 4\times10^{11}\,\text{N/m}^2$

[27 July, 2022 (Shift-II)]

23. (13)



$\tan r = 2$

$\sin i = \frac{20}{10\sqrt{13}}$

Using snell's law

$\mu_{liquid} = \sqrt{\frac{13}{5}}$

[3 Sep, 2020 (Shift-I)]

24. (50)

$F = qvB$

$qE = qvB$

$\frac{\Delta V}{d} = vB$

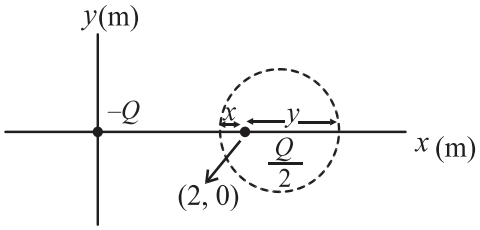
$\Delta V = Bvd$

$\Delta V = \frac{5}{10}\times1\times\frac{10}{100}$

$\Delta V = 50\,\text{mV}$

[11 April, 2023 (Shift-II)]

25. (4)



$\frac{KQ}{2-x} = \frac{KQ}{2x}$

$x = \frac{2}{3}\,\text{m}$

$\frac{KQ}{2+y} = \frac{KQ}{2y}$

$y = 2\,\text{m}$

$x + y = \frac{8}{3}\,\text{m}$

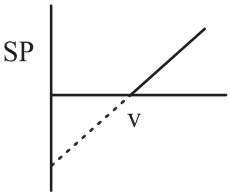
$R = \frac{4}{3}\,\text{m}$

[8 April, 2023]

SECTION-II (CHEMISTRY)

26. **(2)**

K.E = hv-hv$_0$, so (2) is not correct.



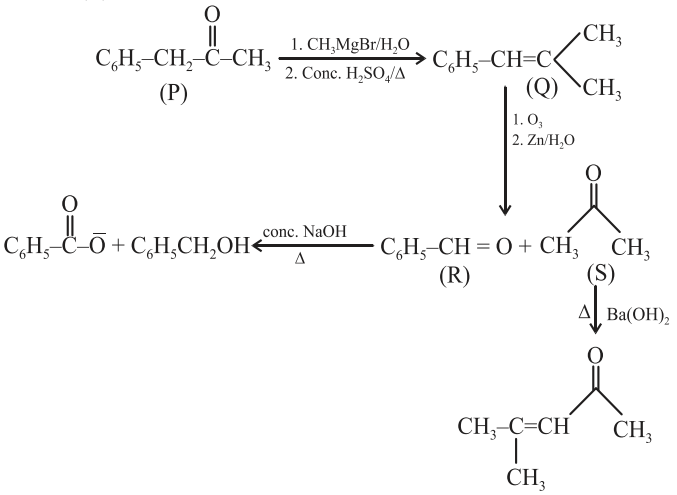
[10 Jan, 2019 (Shift-I)]

27. **(4)**

The titration curve of strong acid HCl vs strong base NaOH is given in option (4). With the addition of NaOH to HCl, pH of solution increases gradually and then a steep curve is obtained. At neutralisation point pH becomes 7 which is for neutral solution.

[27 July, 2022 (Shift-II)]

28. **(2)**



[3 Sept, 2020 (Shift-II)]

29. **(3)**

Ni^{2+} + Dimethylglyoxime (DMG) $\rightarrow$

$$\underset{\text{(Bright red ppt.)}}{\left[Ni(DMG)_2\right]\downarrow}$$

$Ni^{2+} \rightarrow$ Group IV

4 chelate rings

–2 six-membered & 2 five membered.

[24 June, 2022 (Shift-1)]

30. **(2)**

The standard electrode potential of M^+/M in aqueous solution does not depend on ionisation of a solid metal as for ionisation, solid should be converted to the gaseous state first. Direct ionisation is not possible.

[13 April, 2023 (Shift-I)]

31. **(2)**

$A \xrightarrow{k_1} B \xrightarrow{k_2} C$

If $k_2 > k_1$, then graph in option (2) is correct

[8 April, 2019 (Shift-II)]

32. **(4)**

As the size of cation decreases, covalent character of the compound increases. Zn^{2+} has more polarising power than Fe^{2+}

[12 April, 2023 (Shift-I)]

33. **(3)**

element	Cr	Mn	Fe	Co
M^{3+}/M^{2+}	–0.41	1.57	0.77	1.97

[24 Feb, 2021 (Shift-I)]

34. **(2)**

$[NiCl_4]^{2-} \rightarrow$ Ni has sp^3 hybridization having tetrahedral shape.

$[Fe(CN)_6]^{4-} \rightarrow$ Fe has d^2sp^3 hybridization.

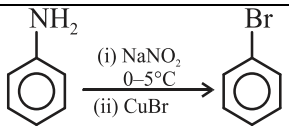
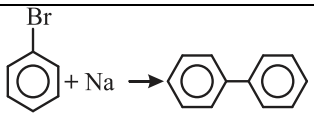
$PF_5 \rightarrow$ P has sp^3d hybridization.

$[Co(F)_6]^{3-} \rightarrow$ Co has sp^3d^2 hybridization.

Here, Co is in +3 oxidation state, F^- will act as a weak field ligand.

[24 June, 2022 (Shift-I)]

35. **(1)**

List-I		List-II	
A		Q	Sandmeyer Reaction
B		P	Fittig reaction
C	EtCl + AgF $\rightarrow$ Et–F + AgCl	S	Swarts Reaction
D	MeBr + NaI $\rightarrow$ MeI + NaBr	R	Finkelstein reaction

[30 Jan, 2023 (Shift-I)]

36. **(1)**

Z = 13, 'Al'

$$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O(\ell)$$

$$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O(\ell)$$

With 1 mol Al_2O_3,

$n_{HCl} : n_{NaOH}$

6 : 2

3 : 1

[9 Jan, 2020 (Shift-II)]

37. **(1)**

Due to extremely small size, bare proton does not exist independently, it is highly reactive.

$$H^+ + H_2O \longrightarrow H_3O^+ \xrightarrow{H_2O} H_5O_2^+ \xrightarrow{H_2O} H_7O_3^+$$

[6 April, 2023 (Shift-I)]

38. **(3)**

Z = 119

Ununennium (Uue)

[12 Jan, 2019 (Shift-I)]

39. **(3)**

$NO_2^- < NO_2 < NO_2^+$

$F_2O < H_2O < Cl_2O$

$AsH_3 < PH_3 < NH_3$

[5 Sept, 2020 (Shift-II)]

40. **(3)**

Non-metal displacement (redox)

reaction include H_2 or O_2 displacement (A and C)

[6 April, 2023 (Shift-I)]

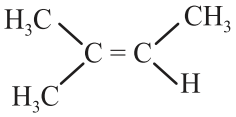
41. **(4)**

All the given statements except 4th regarding group -13 are correct

Tl^{3+} is unstable in aqueous solution and is a strong oxidant. $Tl^+ > Tl^{3+}$ (Stability, due to inert pair effect)

[9 April, 2019 (Shift-II)]

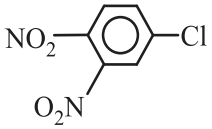
42. **(2)**

 does not show geometrical isomerism

[5 Sept, 2020 (Shift-II)]

43. **(2)**

As the number of $-NO_2$ groups increases rate of nucleophilic aromatic substitution reactions increases.

Hence compound  will have highest rate in nucleophilic aromatic substitution and it will be least reactive towards EAS.

[25 Jan, 2023 (Shift-I)]

44. **(1)**

Molality is defined as the number of moles of solute per kg solvent. Hence, molality does not contain volume. Hence independent of temperature.

[27 June, 2022 (Shift-I)]

45. **(2)**

(i) $MeBr + NH_3 \rightarrow MeNH_2 \xrightarrow{MeBr}$

$Me_2NH \xrightarrow{MeBr} Me_3N \xrightarrow{MeBr} Me_4\overset{+}{N}Br^-$

(ii) $MeBr + \underset{(excess)}{NH_3} \rightarrow \underset{(Major)}{MeNH_2} + HBr$

[16 March, 2021 (Shift-II)]

46. **(3)**

Biuret test is given by urea, all proteins and peptides having atleast two peptide linkages.

Hence, positive test must be given by urea, glutathione, glycyhistidyllysine.

[25 June, 2022 (Shift-II)]

47. **(50)**

Work done, w = –2 (5V – V) = –8 V atm L

$\Delta U = 0$, In an isothermal process,

According to first law of thermodynamics,

$\Delta U = q + w = 0$

w = –q = –20000 J

$q = -\frac{20000}{100} = -200$ atm L

–8 V = –200

$V = \frac{50}{2} \Rightarrow 2V = 50$

[28 June, 2022 (Shift-I)]

48. **(92)**

$$E = E° - 0.06\log\frac{[H^+][Cl^-]}{[P_{H_2}]^{1/2}}$$

$$0.80 = E° - 0.06\log\frac{10^{-5}\cdot 10^{-5}}{1^{1/2}}$$

$$0.80 = E° - 0.06\log 10^{-10}$$

$$E° = 0.80 - 0.06\times 10$$

$$E° = 0.80 - 0.60$$

$$E^\circ_{Cl^-/AgCl/Ag} = 0.20\,V$$

When pH of HCl = 6

Then $[H^+] = [Cl^-] = 10^{-6}$ M

$E = 0.20 - 0.06 \times \log(10^{-12})$

E = 0.92 V

X = 0.92

[10 Jan, 2019 (Shift-II)]

49. **(4)**

$XY_2(g)$

$t_{90\%}$ = 12 min

$$\frac{2.303}{k}\log\left(\frac{100}{100-90}\right)=12$$

$$k=\frac{2.303}{12}\min^{-1}$$

$$t_{1/2}=\frac{0.693}{k}=\frac{2.303\log 2}{2.303}\times 12$$

$$t_{1/2}=3.6\,\text{min}$$

For $PQ_2(g)$

$$t_{90\%}=9\,\text{min}$$

$$k=\frac{2.303}{9}\min^{-1}$$

$$t_{1/2}=\frac{2.303\log 2}{2.303}\times 9=2.7\,\text{min}$$

$$\Rightarrow \quad 3\times\frac{\left(t_{\frac{1}{2}}\right)_{XY_2}}{\left(t_{\frac{1}{2}}\right)_{PQ_2}}=3\times\frac{3.6}{2.7}=4$$

[26 July, 2022 (Shift-II)]

50. (3)

For $[Co(H_2O)_6]^{2+}$

$Co^{2+} \Rightarrow 3d^7$

$\Rightarrow \quad t_{2g}^5 e_g^2$

$\Rightarrow \quad CFSE = (-0.4 \times 5 + 0.6 \times 2)\,\Delta_0$

$= -0.8\,\Delta_0$

For $\left[CoCl_4\right]^{2-}$

$Co^{2+} - 3d^7 \Rightarrow e^4 t_2^3$

$CFSE = \left[4\cdot(0.6)+3(-0.4)\right]\Delta_t$

$= 1.2\Delta_t$

$$\Delta_t=\frac{4}{9}\Delta_0$$

$$2\times\frac{(CFSE)_{aqua}}{(CFSE)_{Chloro}}=\frac{|-0.8\Delta_0|}{1.2\times\frac{4}{9}\Delta_0}\times 2$$

$$=\frac{8\times 9\times 2}{12\times 4}=3$$

[31 Jan, 2023 (Shift-II)]

SECTION-III (MATHEMATICS)

51. (1)

$4(3k-3)^2 - 4(8k^2 - 63) < 0$

$9k^2 - 18k + 9 - 8k + 63 - 8k^2 < 0$

$k^2 - 18k + 72 < 0$

$(k-12)(k-6) < 0$

[25 Feb, 2021 (Shift-I)]

52. (2)

Total number of ways = 6^4

Favourable outcomes = ${}^6P_4$

$\therefore$ Probability = $\frac{5}{18}$

Given, probability = $\frac{p}{q}$

$\therefore \quad p = 5, q = 18$

Hence, $q - p = 13$

[6 April, 2023 (Shift-II)]

53. (1)

$e^{4x} + 8e^{3x} + 10e^{2x} - 8e^x + 1 = 0$

Let $e^x = y$

Now $y^4 + 8y^3 + 10y^2 - 8y + 1 = 0$

Dividing equation by y^2

$$y^2+8y+10-\frac{8}{y}+\frac{1}{y^2}=0$$

$$y^2+\frac{1}{y^2}+8\left(y-\frac{1}{y}\right)+10=0$$

$$\left(y-\frac{1}{y}\right)^2+2+8\left(y-\frac{1}{y}\right)+10=0$$

Let $y-\frac{1}{y}=p$

$p^2 + 8p + 12 = 0$

$(p+6)(p+2) = 0$

$p = -6$ or $p = -2$

$y=-3\pm\sqrt{10}$

$e^x=-3\pm\sqrt{10},\ e^x=-1\pm\sqrt{2}$

$x=\log\left(-3+\sqrt{10}\right)$

$x=\log\left(\sqrt{2}-1\right)$

Hence, two solution and both are negative

[31 Jan, 2023 (Shift-II)]

54. (3)

Given, $\log_{7^{\frac{1}{2}}} x+\log_{7^{\frac{1}{3}}} x+\log_{7^{\frac{1}{4}}} x+\ldots\ldots+\log_{7^{\frac{1}{12}}} x$ =270

$\Rightarrow$ $2\log_7 x + 3\log_7 x + 4\log_7 x + \ldots + 12\log_7 x = 270$

$\Rightarrow$ $\log_7 x + \log_7 x + 3\log_7 x + 4\log_7 x + \ldots + 12\log_7 x - \log_7 x = 270$

$\Rightarrow$ $(1 + 2 + 3\ldots + \ldots + 13)\log_7 x - \log_7 x = 270$

$\Rightarrow$ $x = 343$

[20 July, 2021 (Shift-II)]

55. (1)

$$A=\begin{bmatrix}4 & -1\\ 12 & -3\end{bmatrix}$$

$$B=\begin{bmatrix}3 & -6\\ 1 & -2\end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} = A$$

$\Rightarrow \ ....= A^3 = A^2 = A \Rightarrow A^n = A$

$$B^2 = \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix} = B$$

$\Rightarrow \ ...B^3 = B^2 = B \Rightarrow \ B^n = B$

Now, $nA^n + mB^m = I \Rightarrow \ nA + mB = I$

$4n + 3m = 1$

$-n - 6m = 0$

$$m = -\frac{1}{21}$$

$$n = \frac{2}{7}$$

[25 June, 2022 (Shift-II)]

56. (3)

$$z = \frac{(5 + i\sin\theta)}{(2 - i\cos\theta)} \times \frac{(2 + i\cos\theta)}{(2 + i\cos\theta)}$$

As z is purely real

$\Rightarrow \ 5\cos\theta + 2\sin\theta = \theta \ \Rightarrow \tan\theta = -\frac{5}{2}$

$$\arg(\sin\theta + i\cos\theta) = \pi + \tan^{-1}\left(\frac{\cos\theta}{\sin\theta}\right)$$

$$= \pi + \tan^{-1}\left(-\frac{2}{5}\right) = \pi - \tan^{-1}\left(\frac{2}{5}\right)$$

[17 Jan, 2023 (Shift-II)]

57. (3)

$x_1 + x_2 + x_3 = 24$

$x_i \geq 2$

$^{20}C_2$

[6 April, 2023 (Shift-I)]

58. (2)

$$f(x) = \ln(4x^2 + 15x + 9) + \cos^{-1}(4x + 3) + \sin^{-1}\left(\frac{10x + 6}{3}\right)$$

For $4x^2 + 15x + 9 > 0$

$(4x + 3)(x + 3) > 0$

$$x \in (-\infty, -3) \cup \left(-\frac{3}{4}, \infty\right)$$

For $4x + 3 \in [-1, 1]$

$x \in [-1, -1/2]$

For $\frac{10x + 6}{3} \in [-1, 1]$

$$x \in \left[-\frac{9}{10}, -\frac{3}{10}\right]$$

Hence, domain of the function

$$x \in \left(-\frac{3}{4}, -\frac{1}{2}\right] \Rightarrow \alpha = -\frac{3}{4}, \beta = -\frac{1}{2}$$

$$\therefore \ 48|\alpha - \beta| = 48\left|\frac{-3}{4} + \frac{1}{2}\right| = 12$$

[15 April, 2023 (Shift-I)]

59. (3)

$S = 0.(^{40}C_0)^2 + 1.(^{40}C_1)^2 + 2.(^{40}C_2)^2 + \ldots\ldots + 40.(^{40}C_{40})^2$ .....(i)

$S = 40.(^{40}C_0)^2 + 39.(^{40}C_1)^2 + 38.(^{40}C_2)^2 + \ldots\ldots + 0.(^{40}C_0)^2$ .....(ii)

On adding both equation (i) and (ii),we get

$2S = 40.(^{40}C_0{}^2 + 40C_1{}^2 + \ldots\ldots + {}^{40}C_{40}{}^2)$

$$S = 20.{}^{80}C_{40} = 20.\frac{80!}{(40!)^2} \quad \left[\because \ {}^nC_0^2 + {}^nC_1^2 + \ldots + {}^nC_n^2 = {}^{2n}C_n\right]$$

So, $\frac{20.80!}{(40!)^2} = \frac{\alpha.80!}{(40!)^2}$

$\Rightarrow \ \alpha = 20$

[24 Jan, 2023 (Shift-II)]

60. (3)

$$\alpha, \beta = \frac{-\sqrt{8} \pm \sqrt{8 - 16}}{2} = \frac{\sqrt{8}}{2}(-1 \pm i)$$

$$= 2\left(\frac{-1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i\right)$$

$$= 2e^{\pm\frac{3\pi i}{4}}$$

Required expression

$$= \frac{(2)^{25}\left(2\cos\frac{75\pi}{4}\right) + (2)^{14}\left(2\cos\frac{42\pi}{4}\right)}{(2)^{17}\left(2\cos\frac{51\pi}{4}\right) + (2)^{10}\left(2\cos\frac{30\pi}{4}\right)}$$

$$= (2)^8 = 256$$

[12 April, 2023 (Shift-I)]

61. (3)

No. of ways of select and arrange x_1, x_2, x_3, x_4, x_5 from 1, 2, 3……18

$n(s) = {}^{18}C_5$

$x_1 \quad (x_2) \quad x_3 \quad (x_4) \quad x_5$

$\quad\quad 9 \quad\quad\quad 12$

$n(E) = {}^8C_1 \times {}^2C_1 \times {}^6C_1$

$$P(E) = \frac{8 \times 2 \times 6}{2^3 \times 3^2 \times 7 \times 17}$$

$$\Rightarrow \frac{4}{357}$$

[27 June, 2023 (Shift-I)]

62. (2)

$z = x + iy$

Given $\frac{2z - 3}{2z + 1}$ is purely imaginary

$$\frac{2z - 3}{2z + 1} + \frac{2\bar{z} - 3}{2\bar{z} + 1} = 0$$

$$8z\bar{z} - 4(z + \bar{z}) = 6$$

$$8(x^2 + y^2) - 8x = 6$$

$$8(y^4 + y^2) + 8y^2 = 6$$

$$y^4 + 2y^2 = \frac{3}{4}$$

[10 April, 2023 (Shift-I)]

63. (2)

$$Y=\alpha\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}-\beta\begin{bmatrix}0&1&0\\0&0&1\\0&0&0\end{bmatrix}+\gamma\begin{bmatrix}0&0&1\\0&0&0\\0&0&0\end{bmatrix}$$

$$YY^{-1}=I\Rightarrow\begin{bmatrix}\alpha&-\beta&\gamma\\0&\alpha&-\beta\\0&0&\alpha\end{bmatrix}\begin{bmatrix}\frac{1}{5}&-\frac{2}{5}&\frac{1}{5}\\0&\frac{1}{5}&-\frac{2}{5}\\0&0&\frac{1}{5}\end{bmatrix}=I$$

$\alpha = 5$

$-\frac{2\alpha}{5}-\frac{\beta}{5}=0$

$\beta = -10$

$\frac{\alpha}{5}+\frac{2\beta}{5}+\frac{\gamma}{5}=0$

$\gamma = 15$

[26 June, 2022 (Shift-II)]

64. (4)

We are given that, $\tan\alpha=1, \sec\beta=\frac{-5}{4}$,

$\Rightarrow \quad \cos\beta=-\frac{4}{5}, \Rightarrow \tan\beta=-\frac{3}{4}$, and $\tan\alpha = 1$

$\therefore \tan(\alpha+\beta)=\frac{1-\frac{3}{4}}{1+\frac{4}{3}\times1}=\frac{1}{7}$

$\tan(\alpha+\beta)=\frac{1}{7}$ lies in Ist quadrant

$\because \left[\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\cdot\tan\beta}\right]$

[28 June, 2022 (Shift-II)]

65. (4)

$$\sin^{-1}\left(\frac{3}{10}\sin\left(\cot^{-1}\left(\frac{4}{3}\right)\right)+\frac{2}{5}\cos\left(\cos^{-1}\left(\frac{4}{3}\right)\right)\right)$$

$$=\sin^{-1}\left(\frac{3}{10}.\frac{3}{5}+\frac{2}{5}.\frac{4}{5}\right)=\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}$$

[26 June, 2022 (Shift-II3)]

66. (1)

$^{10}C_4 = 210$

[12 Jan, 2019 (Shift-I)]

67. (3)

$$\lim_{n\to\infty}\left(1+\frac{1^2-2^2+3^2-4^2.....-n^2}{n^3}\right)^n$$

$$=e^{\lim_{n\to\infty}\frac{-(n)(n+1)n}{2n^3}}=e^{-\frac{1}{2}}$$

limit is in the form of 1^∞

[25 Feb, 2021 (Shift-I)]

68. (1)

$f(x) = x + 3x^2 + 5x^3 + \ldots\ldots + 17x^9$

$xf(x) = \quad x^2 + 3x^3 + \ldots\ldots\ldots + 17x^{10}$

$f(x)(1-x) = x + 2x^2 + 2x^3 \ldots\ldots 9\text{terms} - 19x^{10}$

$\Rightarrow \quad f(x)\,(1-x)^2$

$= (1-x)x + 2x^2\,(1-x^8) - 17x^{10}(1-x)$

$\Rightarrow \quad f'(x)\,(1-x)^2 - 2\,(1-x)\,f(x)$

$= 1 + 2x - 190\,x^9 + 187\,x^{10}$

$\Rightarrow \quad f'(2) + 2f(2) = 94213 = 43 \times 2191$

[13 April, 2023 (Shift-I)]

69. (1)

$$I(x)=\int\frac{(x+1)e^x}{xe^x(1+xe^x)^4}dx$$

Let $1 + xe^x = t$

$= e^x\,(x+1)\,dx = dt$

$$I(x)=\int\frac{1}{(t-1)t^4}dt=\int\left(\frac{1}{t-1}-\frac{1}{t}-\frac{1}{t^2}-\frac{1}{t^3}-\frac{1}{t^4}\right)dt$$

$$=\ln\left(\frac{t-1}{t}\right)+\frac{1}{t}+\frac{1}{2t^2}+\frac{1}{3t^3}+C$$

$$\ln\left(\frac{xe^x}{xe^x+1}\right)+\frac{1}{\left(xe^x+1\right)}+\frac{1}{2\left(xe^x+1\right)^2}+\frac{1}{3\left(xe^x+1\right)^3}+C$$

[8 April, 2023 (Shift-I)]

70. (1)

$$\because \frac{dy}{dx}=\tan x\left(\frac{3y}{\ln\cos x}+\frac{1}{(\ln(\cos x))^2}\right)$$

$$\Rightarrow \frac{dy}{dx}-\left(\frac{3\tan x}{\ln\cos x}\right)y=\frac{\tan x}{(\ln(\cos x))^2}$$

I.F. $=e^{\int\frac{-3\sin x dx}{\ln\cos x}}$

Lets $\ln\cos x = t \Rightarrow \frac{-\sin x}{\cos x}dx = dt$

$\therefore$ I.F. $=e^{\int\frac{3dt}{t}}=e^{3\ln t}=t^3=(\ln\cos x)^3$

Solution is given by

$$y\cdot(\ln\cos x)^3=\int\left(\frac{\sin x}{\cos x}\cdot\ln\cos x\right)dx+C$$

$$\Rightarrow y(\ln\cos x)^3=\frac{-(\ln\cos x)^2}{2}+C$$

Given is satisfy $y\left(\frac{\pi}{4}\right)=\frac{1}{2\ln\sqrt{2}}$

$$y\cdot\left(\ln\frac{1}{\sqrt{2}}\right)^3=-\frac{\left(\ln\frac{1}{\sqrt{2}}\right)^2}{2}+C$$

$$\Rightarrow \quad \frac{1}{2\ln\sqrt{2}}\times-\ln\sqrt{2}=-\frac{1}{2}+C$$

$$\Rightarrow \quad c=0$$

$$\Rightarrow \quad 2y=-\frac{1}{\ln\cos x}$$

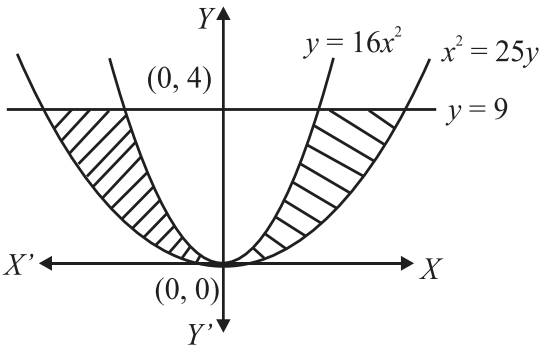
$$\Rightarrow \quad 2y\left(x=\frac{\pi}{6}\right)=-\frac{1}{\ln\cos\frac{\pi}{6}}=-\frac{1}{\ln\frac{\sqrt{3}}{2}}$$

$$=\frac{1}{\ln 2-\ln\sqrt{3}}$$

$$\Rightarrow \quad m=2,\ n=3$$

[8 April, 2023 (Shift-II)]

71. (171)



Required area = $2\int_0^9\left(5\sqrt{y}-\frac{\sqrt{y}}{4}\right)dy$

$$2\int_0^9\frac{19\sqrt{y}}{4}dy$$

[27 July, 2022 (Shift-II)]

72. (8)

$$\int_0^{\pi/8}\left[\sec x+2\left[\tan x+3[\cos x]+12[\sin x]\right]\right]dx$$

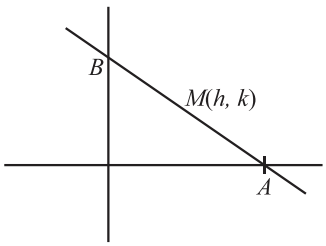
$$=\int_0^{\pi/8}\left[\sec x+2[\tan x]+6[\cos x]+24[\sin x]\right]dx$$

$$=\int_0^{\pi/8}[\sec x]dx+2\int_0^{\pi/8}[\tan x]dx+6\int_0^{\pi/8}[\cos x]dx+24\int_0^{\pi/8}[\sin x]dx$$

$$=\int_0^{\pi/8}[\sec x]dx=\int_0^{\pi/8}dx=\frac{\pi}{8}$$

73. (5)

Let $M(h, k)$ is the mid-point



$x\cos\theta+y\sin\theta=5$

x-intercept = $\frac{5}{\cos\theta}$

y-intercept = $\frac{5}{\sin\theta}$

$$A\left(\frac{5}{\cos\theta},0\right),B\left(0,\frac{5}{\sin\theta}\right)$$

Locus of mid point is M(h, k) and M and $\left(\beta,\frac{5\sqrt{3}}{3}\right)$ also lies on same curve

$$h=\frac{5}{2\cos\theta},k=\frac{5}{2\sin\theta}\Rightarrow\frac{5}{2\sin\theta}=\frac{5\sqrt{3}}{3}$$

$$\Rightarrow\ \sin\theta=\frac{\sqrt{3}}{2}\Rightarrow\theta=\frac{\pi}{3}\Rightarrow\beta=\frac{5}{2\cos\theta}=5$$

[12 April, 2023, (Shift-I)]

74. (1)

$$\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$$

$$\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$$

Projection of $\vec{a}$ on $\vec{b}$ is

$$\vec{a}.\hat{b}=\frac{5+2-6}{3}=\frac{1}{3}$$

[1 Feb, 2023 (Shift-II)]

75. (5)

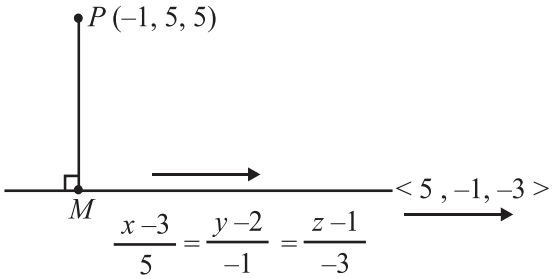
Direction ratio of line $\begin{vmatrix} \hat{i} & \hat{j} & k \\ 1 & 2 & 1 \\ 0 & -3 & 1 \end{vmatrix}$

$= \hat{i}(5) - \hat{j}(1) - \hat{k}(3)$

$= 5\hat{i} - \hat{j} - 3\hat{k}$

Equation of line

$$\frac{x-3}{5} = \frac{y-2}{-1} = \frac{z-1}{-3}$$



$$\frac{x-3}{5} = \frac{y-2}{-1} = \frac{z-1}{-3}$$

$M\,(5k + 3, -k + 2, -3k + 1)$

$\overrightarrow{PM} \perp (5\hat{i} - \hat{j} - 3\hat{k})$

$\Rightarrow \quad k = -1$

Point $M = (-2, 3, 4) \equiv (\alpha, \beta, \gamma)$

$\alpha + \beta - \gamma = 5$

$\alpha = -2, \beta = 3, \gamma = 4$

[24 Jan, 2023 (Shift-II)]

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